

Critical Thinking Assignment 1: Labs Lesson 1 and 2

Alexander Ricciardi

Colorado State University Global

IT315-2: Introduction to Networks

Dr. Sheryl Drake

December 15, 2024

Critical Thinking Assignment 1: Labs Lesson 1 and 2

This documentation is part of the Critical Thinking 1 Assignment from ITS315: Introduction to Networks at Colorado State University Global.

The Assignment Direction:

Module #1: uCertify Lab Simulations

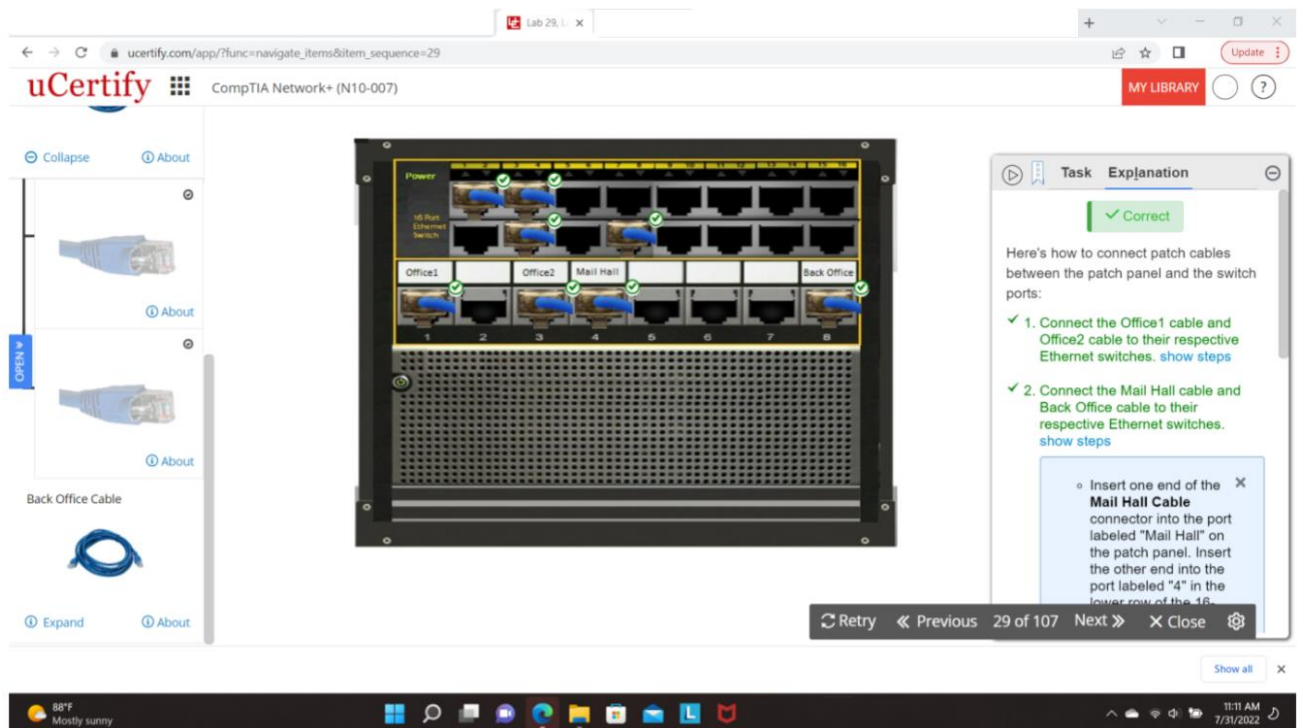
For this assignment, you will complete multiple lab simulations. Activities include identifying network components, network topologies, OSI layers, and TCP/IP protocol layers. You will take a screenshot upon completion of each lab and include the screenshots in the submitted assignment.

Access uCertify and login, go to Labs, and complete the tasks in the following lab simulations:

- 1.1.1 Identifying network components
- 1.2.1 Identifying network categories
- 1.3.2 Identifying network topologies
- 2.1.1 Identifying OSI layers
- 2.1.3 Identifying TCP/IP protocol layers

After completing the task, click Submit >> Evaluate >> Record my answer to record your answer. Take a screenshot of each of the labs and paste the screenshot into a Word document. The document should have a title page that includes your name, date, school name, section, course name, and instructor name. Submit the assignment in Canvas.

Please ensure your screenshot includes your name, date, and timestamp as shown in the image below.



Screenshots

Figure 1
1.1.1 Identifying network components

The screenshot shows a web browser window with the URL `seu-csuglobal.ucertify.com/app/?func=navigate_items&item_sequence=1`. The page header includes the CSU GLOBAL logo and course information: **24WB-ITS315-2** (COURSE - CompTIA Network+ (N10-007)). The interface features a navigation bar with 'MY LIBRARY', 'AR', and a question mark icon. Below the header, there are buttons for 'Correct Answer', 'Compare', and 'Your Answer'. The main content area displays a matching exercise with two columns: 'Network Component' and 'Description'. The 'Network Component' column lists: 1. Router, 2. Hub, 3. Media, 4. Server, 5. Switch. The 'Description' column lists: A. A layer 2 device that filters and forwards packets between LAN segments, B. A layer 3 device that forwards data packets along networks, C. A layer 1 device that connects segments of a LAN and contains multiple ports, D. A program, or machine, that waits for incoming requests, E. A copper cabling or a fiber optic cabling that interconnects devices. Green arrows indicate the correct matches: Router to B, Hub to C, Media to E, Server to D, and Switch to A. On the right, an 'Explanation' panel shows a 'Correct' status and a list of definitions for network components. At the bottom, there are buttons for 'RESET', 'RETRY', 'PREVIOUS', '1 of 107', 'NEXT', 'SETTINGS', and 'CLOSE'. The Windows taskbar at the bottom shows the time as 12:14 PM on 12/13/2024.

Network Component	Description
1. Router	B. A layer 3 device that forwards data packets along networks
2. Hub	C. A layer 1 device that connects segments of a LAN and contains multiple ports
3. Media	E. A copper cabling or a fiber optic cabling that interconnects devices
4. Server	D. A program, or machine, that waits for incoming requests
5. Switch	A. A layer 2 device that filters and forwards packets between LAN segments

Types of network components are described below:

- **Hub:** A layer 1 device that connects segments of a LAN and contains multiple ports
- **Switch:** A layer 2 device that filters and forwards packets between LAN segments
- **Router:** A layer 3 device that forwards data packets along networks
- **Server:** A program, or machine, that waits for incoming requests
- **Media:** A copper cabling or a fiber optic cabling that interconnects devices

Lesson
Computer Network Fundamentals
Defining a Network

Figure 2
1.2.1 Identifying network categories

The screenshot shows a web browser window with the URL `seu-csuglobal.ucertify.com/app/?func=navigate_items&item_sequence=2`. The page header includes the CSU GLOBAL logo and course information: **24WB-ITS315-2** (COURSE - CompTIA Network+ (N10-007)). The interface features a navigation bar with 'MY LIBRARY', 'AR', and a question mark icon. Below the header, there are buttons for 'Correct Answer', 'Compare', and 'Your Answer'. The main content area displays a matching exercise with two columns: 'Network Category' and 'Description'. The 'Network Category' column lists: 1. LAN, 2. CAN, 3. PAN, 4. MAN, 5. WAN. The 'Description' column lists: A. Interconnects network components within a local area, B. Interconnects network components that are geographically separated, C. Interconnects locations scattered throughout a metropolitan area, D. Interconnection of local-area networks within a limited geographical space, E. A network whose scale is very small, like a connection between a PC and a digital camera via USB. Green arrows indicate the correct matches: LAN to A, CAN to D, PAN to E, MAN to C, and WAN to B. On the right, an 'Explanation' panel shows a 'Correct' status and a list of definitions for network categories. At the bottom, there are buttons for 'RESET', 'RETRY', 'PREVIOUS', '2 of 107', 'NEXT', 'SETTINGS', and 'CLOSE'. The Windows taskbar at the bottom shows the time as 12:21 PM on 12/13/2024.

Network Category	Description
1. LAN	A. Interconnects network components within a local area
2. CAN	D. Interconnection of local-area networks within a limited geographical space
3. PAN	E. A network whose scale is very small, like a connection between a PC and a digital camera via USB
4. MAN	C. Interconnects locations scattered throughout a metropolitan area
5. WAN	B. Interconnects network components that are geographically separated

Types of network categories are described below:

- **WAN (Wide Area Network):** Interconnects network components that are geographically separated
- **MAN (Metropolitan Area Network):** Interconnects locations scattered throughout a metropolitan area
- **LAN (Local Area Network):** Interconnects network components within a local area
- **PAN (Personal Area Network):** A network whose scale is very small, like a connection between a PC and a digital camera via USB
- **CAN (Campus Area Network):** Interconnection of local-area networks within a limited geographical space

Lesson
Computer Network Fundamentals
Networks Defined by Geography

Figure 3
1.3.2 Identifying network topologies

Correct Answer **Compare** **Your Answer**

Topology

1. Ring
2. Bus
3. Mesh
4. Star

Image

A.

B.

C.

D.

Activity **Explanation**

✓ **Correct**

Network topology refers to layout of a network and how different nodes in a network are connected to each other and how they communicate. Here are the network topologies:

- ✓ Star Topology
- ✓ Mesh Topology
- ✓ Bus Topology
- ✓ Ring Topology

Mesh Topology **Star Topology**

Bus Topology **Ring Topology**

RESET RETRY PREVIOUS 4 of 107 NEXT SETTINGS CLOSE

12:25 PM 12/13/2024

Figure 4
2.1.1 Identifying OSI layers

Correct Answer **Compare** **Your Answer**

Layer

1. Physical
2. Session
3. Transport
4. Application
5. Network
6. Data Link
7. Presentation

Feature

A. Logical addressing

B. Windowing

C. Synchronizing bits

D. Synchronizing transmissions

E. Tearing down a session

F. Service advertisement

G. Encryption

Activity **Explanation**

✓ **Correct**

Here are the features of the OSI model layers:

- ✓ **Physical Layer:**
 - ✓ Synchronizing bits
 - ✓ How bits are represented on the medium
 - ✓ Wiring standards for connectors and jacks
 - ✓ Physical topology
 - ✓ Bandwidth usage
 - ✓ Multiplexing strategy
- ✓ **Data Link Layer:**
 - ✓ Synchronizing transmissions
 - ✓ Physical addressing
 - ✓ Logical topology
 - ✓ Method of transmitting on the media
 - ✓ Connection services
- ✓ **Network Layer:**
 - ✓ Logical addressing
 - ✓ Switching
 - ✓ Route discovery and selection
 - ✓ Connection services
 - ✓ Bandwidth usage
 - ✓ Multiplexing strategy

RESET RETRY PREVIOUS 8 of 107 NEXT SETTINGS CLOSE

1:45 PM 12/13/2024

Figure 5
 2.1.3 Identifying TCP/IP protocol layers

The screenshot displays a web-based quiz interface for the TCP/IP protocol layers. The interface is divided into several sections:

- Header:** The top of the page shows the URL `seu-csuglobal.ucertify.com/app/?func=navigate_items&item_sequence=10` and various browser tabs. The course information is **CSU GLOBAL** 24WB-ITS315-2, COURSE - CompTIA Network+ (N10-007).
- Navigation:** A sidebar on the left contains a blue button labeled **ITEM LIST**.
- Quiz Interface:**
 - At the top, there are two tabs: **Correct Answer** and **Your Answer**.
 - The **Correct Answer** tab displays a diagram titled **TCP/IP Layers** showing four layers in a vertical stack, each with a green checkmark:
 - Application Layer (orange box)
 - Transport Layer (green box)
 - Internet Layer (yellow box)
 - Network Access Layer (blue box)
 - To the right of the diagram, there are four light green boxes labeled: **Transport Layer**, **Internet Layer**, **Network Access Layer**, and **Application Layer**.
- Explanation Panel:** On the right side, there is a panel titled **Activity Explanation** with a green checkmark and the word **Correct**.
 - The text states: "The four architectural layers in the TCP/IP model are described in the order below:"
 - It lists four points:
 - ✓ 1. **Application layer:** Provides access to network resources
 - ✓ 2. **Transport layer:** Responsible for preparing data to be transported across the network
 - ✓ 3. **Internet layer:** Responsible for logical addressing (such as IP Addresses) and routing
 - ✓ 4. **Network Access layer:** Translates logical network address into physical machine address
 - Below the text is a diagram showing the four layers in a vertical stack: **Application Layer** (orange), **Transport Layer** (green), **Internet Layer** (yellow), and **Network Access Layer** (blue).
 - The caption below the diagram is **Figure: TCP/IP Protocol Layers**.
- Footer:** At the bottom, there is a navigation bar with buttons: **RESET**, **RETRY**, **PREVIOUS**, **10 of 107**, **NEXT**, **SETTINGS**, and **CLOSE**. The system clock shows **2:10 PM 12/13/2024**.

Figures 1 through 5 show that all the lab questions were answered correctly.